

**Draft IEEE Standard for
Local and Metropolitan Area Networks—**

Port-Based Network Access Control—Corrigendum 1

Prepared by the
IEEE 802.1 Maintenance Task Group

Developed by the
LAN/MAN Standards Committee
of the IEEE Computer Society

This and the following cover pages are not part of the draft. They provide revision and other information for IEEE 802.1 Working Group members and will be updated as convenient. New participants: Please read these cover pages, they contain information that should help you contribute effectively to this standards development project. The [Introduction to the current draft](#) should be useful to all readers.

The text proper of this draft begins with the [Title page](#).

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1 Introduction to the current draft¹

² This document is a corrigendum to IEEE Std 802.1X-2020 in order to update the attached YANG modules to
³ support the ability to extend the interfaces used.

⁴ These cover pages also include an introduction to [Participation in 802.1 standards development](#), a summary
⁵ of the [PAR \(Project Authorization Request\)](#), and a general discussion of [Draft development](#).

6 YANG modules

⁷ The YANG modules updated by this corrigendum are attached to the draft pdf as plain text (UTF-8) .yang
⁸ files.

⁹ As noted above, the following YANG modules in this draft have been updated as a result of maintenance
¹⁰ activity not addressed by the included amendments:

¹¹ — Added derived-from-or-self to the must statement for the augment of the interfaces container.

¹² A Revision statement has been added to each (if any) of these modules indicating that they are “TO BE
¹³ Published ..” as part of the Revision, together with an indication of the change. The “TO BE ” is to be removed
¹⁴ and the revision date updated at pre-publication time [the revision statements in a published standard reflects
¹⁵ only the historical derivation of each published (i.e. standardized) module version from its published
¹⁶ predecessors. The description of the change from the prior published version should be retained. Note that
¹⁷ these updated modules are not ‘the standard’ at this time.

18 Sources

¹⁹ This draft has been prepared from a set of Framemaker files. For a description of the use of conditional text
²⁰ and other FrameMaker and IEEE Std 802.1X Style considerations applicable to this draft see the
²¹ EDITOR-PLEASE-READ-ME file in the FrameMaker book(s) used to generate this draft.

¹The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor’s discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

⁷ **This page intentionally left blank**

⁷ **This page intentionally left blank**

⁷ **This page intentionally left blank**

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2 All participants in IEEE 802.1 activities should be aware of the Working Group Policies and Procedures, and
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4 IEEE SA Participation Policy. For information on these policies see 1.ieee802.org/rules/ and the slides
5 presented at the beginning of each of our Working Group and Task Group meeting.

6 The IEEE SA [PAR \(Project Authorization Request\)](#) is summarized in these cover pages and a link provided to
7 the full text. As part of the IEEE 802® process, the PAR of each project is reviewed regularly to ensure its
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12 complete. Readers are urged to devote their valuable time and energy only to comments that materially affect
13 either the technical content of the document or the clarity of that technical content. Comments should not
14 simply state what is wrong, but also what might be done to fix the problem.

15 Full participation in the work of IEEE 802.1 requires attendance at IEEE 802 meetings. Information on 802.1
16 activities, working papers, and email distribution lists etc. can be found on the 802.1 Website:

17 <http://ieee802.org/1/>

18 Use of the email distribution list is not presently restricted to 802.1 members, and the working group has a
19 policy of considering comments from all who are interested and willing to contribute to the development of the
20 draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity
21 in past projects. The email lists exist primarily to allow the members of the working group to develop
22 standards, and are not a general forum. All contributors to the work of 802.1 should familiarize themselves
23 with the IEEE patent policy and anyone using the email distribution list will be assumed to have done so.
24 Information can be found at <http://standards.ieee.org/db/patents/>

25 Comments on this draft may be sent to the 802.1 email exploder, to the Editor, or to the Chair of the 802.1
26 Working Group..

27 Scott Mansfield
28 Editor, IEEE Std 802.1X-2020/Cor1
29 Email: scott.mansfield@ericsson.com

Glenn Parsons
Chair, 802.1 Working Group
Email: glenn.parsons@ericsson.com

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36 referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity.
37 A vote of “Approve” on this draft is also an affirmation by the balloter that the PAR is still valid.

1 PAR (Project Authorization Request)

2 The PAR for this project has been approved by IEEE SASB, September 10, 2025:

3 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/12877>

4 This is a corrigendum PAR so it did not require an IEEE 802 CSD (Criteria for Standards Development).

5 The Scope and Purpose remain unchanged from that for IEEE Std 802.1X-2020.

6 The Need for the Project is as follows:

7 This corrigendum corrects technical errors identified and agreed by the IEEE 802.1 Working Group
8 maintenance activity.

9 Draft development

10 During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically
11 coherent drafts from the resolutions of ballot comments and from the other discussions that take place in the
12 working group meetings. Preparation of drafts often exposes inconsistencies in editor's instructions or
13 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
14 and requests by the editors' for contributions on specific issues will be found in the editors' [Introduction to the](#)
15 [current draft](#) and at appropriate points in the draft.

16 Any text with a Cyan background (as in this sentence) is temporary, with conditional tag 'Editor comment',
17 inserted by the Editors to solicit comment, suggest a future change, or act simply as an aide memoire. Text
18 can be highlighted to be draw it to the readers' attention, using conditional tag 'Editor highlight'. In both these
19 case conditional tagging helps location, and eventual removal, of text or highlighting and can control whether
20 or not it is displayed.

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22 working group drafts, are part of the audit trail of the development of the standard and are available, along
23 with all the revisions of the draft on the 802.1 website (for address see above).

24 During the early stages of draft development the proposed text can be moved around a great deal, and even
25 minor rearrangement can lead to a lot of 'change', not all of which is noteworthy from the point of the reviewer,
26 so the use of automatic change bars is not very effective. In early drafts change bars may be omitted or
27 applied manually, with a view to drawing the readers attention to the most significant areas of change.
28 Readers interested in viewing every change are encouraged to use Adobe Acrobat to compare the document
29 with their selected prior draft. Note that the FrameMaker change bar feature is useless when it comes to
30 indicating changes to Figures.

31 Records of participants in the development of the standard are added after SA Ballot, as part of
32 pre-publication editing by IEEE Staff.

Draft IEEE Standard for Local and Metropolitan Area Networks— Port-Based Network Access Control—Corrigendum 1

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

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Abstract: This corrigendum to IEEE Std 802.1X-2020 corrects the when statement for the augment of the interfaces container in the ieee802-dot1x YANG module to avoid limiting the latter's applicability..

Keywords: access control, authentication, authorization, controlled port, EAP, EAPOL, IEEE 802.1X, key agreement, LANs, local area networks, MACsec, MACsec Key Agreement, MAC security, MAC Service, MANs, metropolitan area networks, MKA, port-based network access control, secure association, security, service access point, uncontrolled port.

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8 **Karen Randall, *Vice Chair, Security Task Group***
9

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10 The following members of the individual balloting committee voted on this standard. Balloters may have
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A.N. Other

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13

1

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4

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5

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10

¹ Introduction

²

This introduction is not part of IEEE Std 802.1X™-2020X/Cor1, IEEE Standard for Local and metropolitan area networks—Port-Based Network Access Control — Corrigendum 1.

³ This corrigendum to IEEE Std 802.1X™ corrects the when statement for the augment of the interfaces
⁴ container in the ieee802-dot1x YANG module to avoid limiting the latter's applicability.

1 **Contents**

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1

2 Draft IEEE Standard for Local and Metropolitan Area Networks— 4 Port-Based Network Access 5 Control—Corrigendum 1

6 [This corrigendum is based on IEEE Std 802.1X™-2020]

7 NOTE—The editing instructions contained in this corrigendum define how to merge the material contained
8 therein into the existing base standard to form the comprehensive standard.

9 The editing instructions are shown in ***bold italics***. Four editing instructions are used: change, delete, insert,
10 and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies
11 the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old
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15 equations by removing the existing figure or equation and replacing it with a new one. Editing instructions,
16 change markings, and this note will not be carried over into future editions because the changes will be
17 incorporated into the base standard.

1 14. YANG Data Model

2 14.5 Definition of the IEEE 802.1X YANG data model ²3.4

3 14.5.2 ieee802-dot1x-types YANG module definition

4 *Delete the YANG module in 14.5.2.*

5 *Insert the following YANG module in 14.5.2:*

```
6 module ieee802-dot1x-types {
7   yang-version "1.1";
8   namespace "urn:ieee:std:802.1X:yang:ieee802-dot1x-types";
9   prefix "dot1x-types";
10
11   organization
12     "Institute of Electrical and Electronics Engineers";
13
14   contact
15     "WG-URL: http://www.ieee802.org/1
16     WG-EMail: stds-802-1-L@ieee.org
17
18     Contact: IEEE 802.1 Working Group Chair
19     Postal: C/O IEEE 802.1 Working Group
20             IEEE Standards Association
21             445 Hoes Lane
22             Piscataway
23             NJ 08854
24             USA
25
26     E-mail: STDS-802-1-L@LISTSERV.IEEE.ORG";
27
28   description
29     "Port-based network access control allows a network administrator
30     to restrict the use of IEEE 802 LAN service access points (ports)
31     to secure communication between authenticated and authorized
32     devices. IEEE Std 802.1X specifies an architecture, functional
33     elements, and protocols that support mutual authentication
34     between the clients of ports attached to the same LAN and secure
35     communication between the ports. The following control allows a
36     port to be reinitialized, terminating (and potentially
37     restarting) authentication exchanges and MKA operation, based on
38     a data model described in a set of YANG modules.";
39
40   revision 2025-11-20 {
41     description
42       "Added statement to indicate YANG version 1.1";
43     reference
44       "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
45   }
46
47   revision 2020-02-18 {
48     description
49       "Updated Contact information.";
50     reference
51       "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
52   }
53
54   revision 2019-05-28 {
55     description
56       "Updates based upon comment resolution on draft
```

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³ An ASCII version of each YANG module is attached to the PDF of this standard and can also be obtained from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

⁴ References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

```

1      D1.0 of P802.1X-Rev.";
2      reference
3      "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
4  }
5
6  /* -----
7   * Type definitions used by dot1X YANG module
8   * -----
9   */
10
11  typedef pae-nid {
12      type string {
13          length "0..100";
14      }
15      description
16          "Network Identity, which is a UTF-8 string identifying a
17          network or network service.";
18      reference
19          "IEEE 802.1X-2020 Clause 3, Clause 10.1, Clause 12.6";
20  }
21
22  typedef pae-session-user-name {
23      type string {
24          length "0..253";
25      }
26      description
27          "Session user name, which is a UTF-8 string, representing the
28          identity of the peer Supplicant.";
29      reference
30          "IEEE 802.1X-2020 Clause 12.5.1";
31  }
32
33  typedef pae-session-id {
34      type string {
35          length "3..253";
36      }
37      description
38          "Session Identifier, which is a UTF-8 string, uniquely
39          identifying the session within the context of the PAE's
40          system.";
41      reference
42          "IEEE 802.1X-2020 Clause 12.5.1";
43  }
44
45  typedef pae-nid-capabilities {
46      type bits {
47          bit eap {
48              position 0;
49              description
50                  "EAP";
51          }
52          bit eapMka {
53              position 1;
54              description
55                  "EAP + MKA";
56          }
57          bit eapMkaMacSec {
58              position 2;
59              description
60                  "EAP + MKA + MACsec";
61          }
62          bit mka {
63              position 3;
64              description
65                  "MKA";
66          }
67          bit mkaMacSec {
68              position 4;
69              description
70                  "MKA + MACsec";
71          }
72          bit higherLayer {

```

```

1      position 5;
2      description
3          "Higher Layer (WebAuth)";
4      }
5      bit higherLayerFallback {
6          position 6;
7          description
8              "Higher Layer Fallback (WebAuth)";
9      }
10     bit vendorSpecific {
11         position 7;
12         description
13             "Vendor specific authentication mechanisms";
14     }
15 }
16 description
17     "Authentication and protection capabilities supported for the
18     NID. Indicates the combinations of authentication and
19     protection capabilities supported for the NID. Any set of these
20     combinations can be supported.";
21 reference
22     "IEEE 802.1X-2020 Clause 10.1, Clause 11.12.3";
23 }
24
25 typedef pae-access-status {
26     type enumeration {
27         enum no-access {
28             description
29                 "Other than to authentication services, and to services
30                 announced as available in the absence of authentication
31                 (unauthenticated).";
32         }
33         enum remedial-access {
34             description
35                 "The access granted is severely limited, possibly to
36                 remedial services.";
37         }
38         enum restricted-access {
39             description
40                 "The Controlled Port is operational, but restrictions have
41                 been applied by the network that can limit access to some
42                 resources.";
43         }
44         enum expected-access {
45             description
46                 "The Controlled Port is operational, and access provided is
47                 as expected for successful authentication and authorization
48                 for the NID.";
49         }
50     }
51     description
52         "Indicates the transmitter's Controlled Port operational status
53         and current level of access resulting from authentication and
54         the consequent authorization controls applied by that port's
55         clients.";
56     reference
57         "IEEE 802.1X-2020 Clause 10.4, Clause 12.5";
58 }
59
60 typedef mka-kn {
61     type uint32;
62     description
63         "Indicates a Key Number (KN) used in MKA. It is assigned by
64         the Key Server (sequentially beginning with 1).";
65     reference
66         "IEEE 802.1X-2020 Clause 9.8, Clause 9.16";
67 }
68
69 typedef mka-an {
70     type uint32;
71     description
72         "A number that is concatenated with a MACsec Secure Channel

```

```

1     Identifier to identify a Secure Association. Indicates an
2     Association Number (AN) assigned by the Key Server for use with
3     the key number for transmission.";
4     reference
5         "IEEE 802.1X-2020 Clause 9.8, Clause 9.16";
6 }
7
8 typedef pae-ckn {
9     type string {
10         length "1..32";
11     }
12     description
13         "Indicates the CAK name to identify the Connectivity
14         Association Key (CAK) which is the root key in the MACsec Key
15         Agreement key hierarchy. All potential members of the CA use
16         the same CKN.";
17     reference
18         "IEEE 802.1X-2020 Clause 9.3.1, Clause 6.2";
19 }
20
21 typedef pae-kmd {
22     type string {
23         length "0..253";
24     }
25     description
26         "A Key Management Domain (KMD). A string of up to 253 UTF-8
27         characters that names the transmitting authenticator's key
28         management domain.";
29     reference
30         "IEEE Clause 12.6";
31 }
32
33 typedef pae-auth-data {
34     type string;
35     description
36         "Authorization data associated with the CAK.";
37     reference
38         "IEEE 802.1X-2020 Clause 9.16";
39 }
40
41 typedef sci-list-entry {
42     type string {
43         length "8";
44     }
45     description
46         "8 octet string, where the first 6 octets represents the MAC
47         Address (in canonical format), and the next 2 octets represents
48         the Port Identifier.";
49     reference
50         "IEEE 802.1AE Clause 7.1.2, Clause 10.7.1";
51 }
52
53 typedef pae-if-index {
54     type int32 {
55         range "1..2147483647";
56     }
57     description
58         "The interface index value represented by this interface.";
59 }
60
61 } // ieee802-dot1x-types
62
63

```

64 14.5.3 ieee802-dot1x YANG module definition

65 *Delete the YANG module in 14.5.3.*

66 *Insert the following YANG module in 14.5.3:*

```

67 module ieee802-dot1x {
68     yang-version "1.1";

```

```

1 namespace "urn:ieee:std:802.1X:yang:ieee802-dot1x";
2 prefix "dot1x";
3
4 import ieee802-types { prefix "ieee"; }
5 import ietf-yang-types { prefix "yang"; }
6 import ietf-interfaces { prefix "if"; }
7 import ietf-system { prefix "sys"; }
8 import iana-if-type { prefix "ianaift"; }
9 import ieee802-dot1x-types { prefix "dot1x-types"; }
10
11 organization
12   "Institute of Electrical and Electronics Engineers";
13
14 contact
15   "WG-URL: http://www.ieee802.org/1
16   WG-EMail: stds-802-1-L@ieee.org
17
18   Contact: IEEE 802.1 Working Group Chair
19   Postal: C/O IEEE 802.1 Working Group
20           IEEE Standards Association
21           445 Hoes Lane
22           Piscataway
23           NJ 08854
24           USA
25
26   E-mail: STDS-802-1-L@LISTSERV.IEEE.ORG";
27
28 description
29   "Port-based network access control allows a network administrator
30   to restrict the use of IEEE 802 LAN service access points (ports)
31   to secure communication between authenticated and authorized
32   devices. IEEE Std 802.1X specifies an architecture, functional
33   elements, and protocols that support mutual authentication
34   between the clients of ports attached to the same LAN and secure
35   communication between the ports. The following control allows a
36   port to be reinitialized, terminating (and potentially
37   restarting) authentication exchanges and MKA operation, based on
38   a data model described in a set of YANG modules.";
39
40 revision 2025-11-20 {
41   description
42     "Added derived-from-or-self function to the when statement
43     for the augment of the interfaces container.";
44   reference
45     "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
46 }
47
48 revision 2020-02-18 {
49   description
50     "Updated Contact information.";
51   reference
52     "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
53 }
54
55 revision 2019-06-12 {
56   description
57     "Updates based on comment resolution of the WG ballot of
58     P802.1X-Rev/D1.0.";
59   reference
60     "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
61 }
62
63 grouping nid-group {
64   description
65     "The PAE NID Group configuration and operational information.";
66   list pae-nid-group {
67     key "nid";
68     description
69       "A list that contains the configuration and operational
70       nodes for the network announcement information for the
71       Logon Process.";
72     leaf nid {

```

```

1      type dot1x-types:pae-nid;
2      description
3          "Identification of the network or network service.";
4      reference
5          "IEEE 802.1X-2020 Clause 12.5";
6  }
7  leaf use-eap {
8      type enumeration {
9          enum never {
10             description
11                 "Never.";
12             }
13          enum immediate {
14             description
15                 "Immediately, concurrently with the use of MKA with any
16                 cached CAK(s).";
17             }
18          enum mka-fail {
19             description
20                 "Not until MKA has failed, if a prior CAK has been
21                 cached.";
22             }
23          }
24      default "immediate";
25      description
26          "Determines when the Logon Process will initiate EAP, if
27          the Supplicant and or Authenticator are enabled, and takes
28          one of the above values.";
29      reference
30          "IEEE 802.1X-2020 Clause 12.5";
31  }
32  leaf unauth-allowed {
33      type enumeration {
34          enum never {
35             description
36                 "Never.";
37             }
38          enum immediate {
39             description
40                 "Immediately, independently of any current or future
41                 attempts to authenticate using the PAE or MKA.";
42             }
43          enum auth-fail {
44             description
45                 "Not until an attempt has been made to authenticate
46                 using EAP, unless neither the supplicant nor the
47                 authenticator is enabled, and MKA has attempted to use
48                 any cached CAK (unless the KaY is not enabled).";
49             }
50          }
51      default "immediate";
52      description
53          "Determines when the Logon Process will tell the CP state
54          machine to provide unauthenticated connectivity, and takes
55          one of the above values.";
56      reference
57          "IEEE 802.1X-2020 Clause 12.5";
58  }
59  leaf unsecure-allowed {
60      type enumeration {
61          enum never {
62             description
63                 "Never.";
64             }
65          enum immediate {
66             description
67                 "Immediately, to provide connectivity concurrently with
68                 the use of MKA with any CAK acquired through EAP.";
69             }
70          enum mka-fail {
71             description
72                 "Not until MKA has failed, or is not enabled.";

```



```

1      }
2      enum mka-server {
3          description
4              "Only if directed by the MKA server.";
5      }
6  }
7  default "immediate";
8  description
9      "Determines when the Logon Process will tell the CP state
10     machine to provide authenticated but unsecured
11     connectivity, takes one of the above values.";
12  reference
13      "IEEE 802.1X-2020 Clause 12.5";
14  }
15  leaf unauthenticated-access {
16      type enumeration {
17          enum no-access {
18              description
19                  "Other than to authentication services.";
20          }
21          enum fallback-access {
22              description
23                  "Limited access can be provided after authentication
24                  failure.";
25          }
26          enum limited-access {
27              description
28                  "Immediate limited access is available without
29                  authentication.";
30          }
31          enum open-access {
32              description
33                  "Immediate access is available without
34                  authentication.";
35          }
36      }
37      default "no-access";
38      description
39          "Unauthenticated access capabilities provided by the NID.";
40      reference
41          "IEEE 802.1X-2020 Clause 10.1";
42  }
43  leaf access-capabilities {
44      type dot1x-types:pae-nid-capabilities;
45      description
46          "Authentication and protection capabilities supported for
47          the NID.";
48      reference
49          "IEEE 802.1X-2020 Clause 10.1";
50  }
51
52  leaf kmd {
53      type dot1x-types:pae-kmd;
54      config false;
55      description
56          "The Key Management Domain for the NID.";
57      reference
58          "IEEE 802.1X-2020 Clause 10.4";
59  }
60  }
61  }
62
63  grouping port-capabilities {
64      description
65          "Per port PAE feature capabilities.";
66      leaf supp {
67          type boolean;
68          description
69              "Indicates if PACP EAP Supplicant is supported.";
70          reference
71              "IEEE 802.1X-2020 Clause 12.9.2";
72      }

```

```

1  leaf auth {
2      type boolean;
3      description
4          "Indicates if PACP EAP Authenticator is supported.";
5      reference
6          "IEEE 802.1X-2020 Clause 12.9.2";
7  }
8  leaf mka {
9      type boolean;
10     description
11         "Indicates if MKA is supported.";
12     reference
13         "IEEE 802.1X-2020 Clause 12.9.2";
14 }
15 leaf macsec {
16     type boolean;
17     description
18         "Indicates if MACsec on the Controlled port is supported.";
19     reference
20         "IEEE 802.1X-2020 Clause 12.9.2";
21 }
22 leaf announcements {
23     type boolean;
24     description
25         "Indicates if the ability to send EAPOL announcements is
26         supported.";
27     reference
28         "IEEE 802.1X-2020 Clause 12.9.2";
29 }
30 leaf listener {
31     type boolean;
32     description
33         "Indicates if the ability to use received EAPOL
34         announcements is supported.";
35     reference
36         "IEEE 802.1X-2020 Clause 12.9.2";
37 }
38 leaf virtual-ports {
39     type boolean;
40     description
41         "Indicates if virtual ports for a real port is supported.";
42     reference
43         "IEEE 802.1X-2020 Clause 12.9.2";
44 }
45 leaf in-service-upgrades {
46     type boolean;
47     description
48         "Indicates if MKA in-service upgrades is supported.";
49     reference
50         "IEEE 802.1X-2020 Clause 12.9.2";
51 }
52 }
53
54 /* -----
55  * Configuration objects used by 802.1X YANG module
56  * -----
57  */
58 augment "/sys:system" {
59     description
60         "Augment system with 802.1X PAE System specific configuration
61         nodes.";
62     container pae-system {
63         description
64             "Contains all 802.1X PAE System specific related
65             configuration and operational data.";
66         leaf name {
67             type string {
68                 length "1..255";
69             }
70             description
71                 "The name which uniquely identifies the PAE System.";
72         }

```

```

1  leaf system-access-control {
2      type enumeration {
3          enum disabled {
4              description
5                  "Deletes any virtual ports previously instantiated, and
6                  terminates authentication exchanges and MKA
7                  operation.";
8          }
9          enum enabled {
10             description
11                 "Enables PAE system access control.";
12         }
13     }
14     description
15         "Setting this control to disabled deletes any virtual ports
16         previously instantiated, and terminates authentication
17         exchanges and MKA operation. Each real port PAE behaves as
18         if enabledVirtualPorts was clear, the PAEs Supplicant,
19         Authenticator, and KaY as if their enabled controls were
20         clear, and Logon Process(es) as if unauthAllowed was
21         Immediate. Announcements can be transmitted (subject to
22         other controls), both periodically and in response to
23         announcement requests (conveyed by EAPOL-Starts or
24         EAPOL-Announcement-Reqs) but are sent with a single NID
25         Set, with a null NID, and the Access Information TLV (and
26         no other) with an pae-access-status of No Access,
27         accessRequested false, OpenAccess, and no
28         accessCapabilities. The control variable settings for each
29         real port PAE are unaffected, and will be used once
30         systemAccessControl is set to enabled.";
31     reference
32         "IEEE 802.1X-2020 Clause 12.9.1";
33 }
34 leaf system-announcements {
35     type enumeration {
36         enum disabled {
37             description
38                 "Causes each PAE to behave as if enabled were clear
39                 for the PAE's Announcement functionality.";
40         }
41         enum enabled {
42             description
43                 "Enables PAE system announcements.";
44         }
45     }
46     description
47         "Setting this control to Disabled causes each PAE to behave
48         as if enabled were clear for the PAE's Announcement
49         functionality. The independent controls for each PAE apply
50         if systemAnnouncements is Enabled.";
51     reference
52         "IEEE 802.1X-2020 Clause 12.9.1";
53 }
54 leaf eapol-protocol-version {
55     type uint8;
56     config false;
57     description
58         "The EAPOL protocol version for this system.";
59     reference
60         "IEEE 802.1X-2020 Clause 12.9.1, Clause 11.3";
61 }
62 leaf mka-version {
63     type uint8;
64     config false;
65     description
66         "The MKA protocol version for this system.";
67     reference
68         "IEEE 802.1X-2020 Clause 12.9.1, Clause 11.3";
69 }
70 leaf-list pae {
71     type if:interface-ref;
72     config false;

```

```

1      description
2      "List of PAE references.";
3    }
4  }
5 }
6
7 /*
8  * Port Authentication Entity (PAE) Nodes
9  */
10 augment "/if:interfaces/if:interface" {
11   when "derived-from-or-self(if:type,'ianaift:ethernetCsmacd') or " +
12     "derived-from-or-self(if:type,'ianaift:ilan') or " +
13     "derived-from-or-self(if:type,'ianaift:macSecControlledIF') or " +
14     "derived-from-or-self(if:type,'ianaift:ptm') or " +
15     "derived-from-or-self(if:type,'ianaift:bridge')" {
16     description
17       "Applies to the Controlled Port of SecY or PAC shim or
18       Ethernet related Interface.";
19   }
20   description
21     "Augment interface model with PAE configuration and
22     operational nodes.";
23   reference
24     "IEEE 802.1AE Clause 11.7 and IEEE 802.1X-2020 Clause 6.5 and
25     Clause 13.3.2";
26   container pae {
27     description
28       "Contains PAE configuration and operational related nodes.";
29     leaf pae-system {
30       type leafref {
31         path "/sys:system/dot1x:pa-system/dot1x:name";
32       }
33       description
34         "The PAE system that this PAE is a member of.";
35     }
36     leaf vp-enable {
37       when "../port-type = 'real-port' and
38         ../port-capabilities/virtual-ports = 'true'" {
39         description
40           "Applies when port is Real Port and virtual port
41           capabilities are supported.";
42       }
43       type boolean;
44       default "false";
45       description
46         "A real port's PAE may be configured to create virtual
47         ports to support multi-access LANs provided that MKA and
48         MACsec operation is enabled for that port.";
49       reference
50         "IEEE 802.1X-2020 Clause 12.7";
51     }
52     container port-capabilities {
53       description
54         "Per port PAE feature capabilities.";
55       uses port-capabilities;
56     }
57
58     leaf port-name {
59       type if:interface-ref;
60       config false;
61       description
62         "Each PAE is uniquely identified by a port name.";
63     }
64     leaf port-number {
65       type dot1x-types:pa-if-index;
66       config false;
67       description
68         "Each PAE is uniquely identified by a port number. The
69         port number used is unique amongst all port names for the
70         system, and directly or indirectly identifies the
71         Uncontrolled Port that supports the PAE. If the PAE has
72         been dynamically instantiated to support an existing or

```

```

1      potential virtual port, this portNumber, the
2      uncontrolledPortNumber and the controlledPortNumber are
3      allocated by the real ports PAE, and this portNumber is the
4      uncontrolledPortNumber. If the PAE supports a real port,
5      this portNumber is the commonPortNumber for the associated
6      PAC or SecY.";
7      reference
8      "IEEE 802.1X-2020 Clause 12.9.2";
9  }
10 leaf controlled-port-name {
11     type if:interface-ref;
12     config false;
13     description
14         "Each PAE is uniquely identified by a port name.";
15 }
16 leaf controlled-port-number {
17     type dot1x-types:paef-if-index;
18     config false;
19     description
20         "The port for the associated PAC or SecYs Controlled
21         Port.";
22     reference
23         "IEEE 802.1X-2020 Clause 12.9.2";
24 }
25 leaf uncontrolled-port-name {
26     type if:interface-ref;
27     config false;
28     description
29         "The uncontrolled port name reference.";
30 }
31 leaf uncontrolled-port-number {
32     type dot1x-types:paef-if-index;
33     config false;
34     description
35         "The port for the associated PAC or SecYs Uncontrolled
36         Port.";
37     reference
38         "IEEE 802.1X-2020 Clause 12.9.2";
39 }
40 leaf common-port-name {
41     type if:interface-ref;
42     config false;
43     description
44         "The common port name reference.";
45 }
46 leaf common-port-number {
47     type dot1x-types:paef-if-index;
48     config false;
49     description
50         "The port for the associated PAC or SecYs Common Port. All
51         the virtual ports created for a given real port share the
52         same Common Port and commonPortNumber.";
53     reference
54         "IEEE 802.1X-2020 Clause 12.9.2";
55 }
56 leaf port-type {
57     type enumeration {
58         enum real-port {
59             description
60                 "Real Port type.";
61         }
62         enum virtual-port {
63             description
64                 "Virtual Port type.";
65         }
66     }
67     //config false;
68     description
69         "The port type of the PAE.";
70     reference
71         "IEEE 802.1X-2020 Clause 12.9.2";
72 }

```

```

1     container virtual-port {
2         when "../port-capabilities/virtual-ports = 'true'" {
3             description
4                 "Applies when the virtual ports port capability is
5                 supported.";
6         }
7         config false;
8         description
9             "Contains Virtual Port operational state information.";
10        leaf max {
11            when "../port-type = 'real-port'" {
12                description
13                    "Applies when Port is a Real Port.";
14            }
15            type uint32;
16            description
17                "The guaranteed maximum number of virtual ports.";
18            reference
19                "IEEE 802.1X-2020 Clause 12.9.2";
20        }
21        leaf current {
22            when "../port-type = 'real-port'" {
23                description
24                    "Applies when Port is a Real Port.";
25            }
26            type yang:gauge32;
27            description
28                "The current number of virtual ports.";
29            reference
30                "IEEE 802.1X-2020 Clause 12.9.2";
31        }
32        leaf start {
33            when "../port-type = 'virtual-port'" {
34                description
35                    "Applies when Port is a Virtual Port.";
36            }
37            type boolean;
38            description
39                "Set if the virtual port was created by receipt of an
40                EAPOL-Start frame.";
41            reference
42                "IEEE 802.1X-2020 Clause 12.9.7";
43        }
44        leaf peer-address {
45            when "../port-type = 'virtual-port'" {
46                description
47                    "Applies when Port is a Virtual Port.";
48            }
49            type ieee:mac-address;
50            description
51                "The source MAC Address of the EAPOL-Start (if vpStart is
52                set).";
53            reference
54                "IEEE 802.1X-2020 Clause 12.9.7";
55        }
56    }
57
58    container supplicant {
59        when "../port-type = 'real-port' and
60            ../port-capabilities/supp = 'true'" {
61            description
62                "Applies to Real Port when supplicant port capabilities
63                are supported.";
64        }
65        description
66            "Contains the configuration nodes for the Supplicant PAE
67            associated with each port.";
68        leaf held-period {
69            type uint16;
70            units seconds;
71            default "60";
72            description

```

```

1         "The initial value of the timer used to impose a wait
2         period after a failed authentication attempt, before
3         another attempt is permitted.";
4     reference
5         "IEEE 802.1X-2020 Clause 8.6";
6 }
7 leaf retry-max {
8     type uint32;
9     default "2";
10    description
11        "Specifies the maximum number of re-authentication
12        attempts on an authenticator port before port is
13        unauthorized.";
14    reference
15        "IEEE 802.1X-2020 Clause 8.7";
16 }
17
18 leaf enabled {
19     type boolean;
20     config false;
21     description
22         "Set by PACP if the PAE can provide authentication. Will
23         be FALSE if the Port is not enabled, if the functionality
24         provided by the PAE is not available, or not implemented,
25         or the control variable enable has been cleared by
26         management, e.g. because the application scenario
27         authenticates a user and there is no user logged on.";
28     reference
29         "IEEE 802.1X-2020 Clause 8.4";
30 }
31 leaf authenticate {
32     type boolean;
33     config false;
34     description
35         "Set by the PAE client to request authentication, and
36         allows reauthentication while set. Cleared by the client
37         to revoke authentication. To enable authentication the
38         client also needs to clear failed (if set).";
39     reference
40         "IEEE 802.1X-2020 Clause 8.4";
41 }
42 leaf authenticated {
43     type boolean;
44     config false;
45     description
46         "Set by PACP if the PAE is currently authenticated, and
47         cleared if the authentication fails or is revoked.";
48     reference
49         "IEEE 802.1X-2020 Clause 8.4";
50 }
51 leaf failed {
52     type boolean;
53     config false;
54     description
55         "Set by PACP if the authentication has failed or has been
56         terminated. The cause could be a Fail returned by EAP,
57         either immediately or following a reauthentication, an
58         excessive number of attempts to authenticate (either
59         immediately or upon reauthentication), or the client
60         deasserting authenticate. The PACP will clear
61         authenticated as well as setting failed. Any ongoing
62         authentication exchange will be terminated (by the state
63         machines) if enable becomes FALSE and enabled will be
64         cleared, but failed will not be set.";
65     reference
66         "IEEE 802.1X-2020 Clause 8.4";
67 }
68 }
69
70 container authenticator {
71     when "../port-capabilities/auth = 'true'" {
72         description

```

```

1         "Applies when the Authenticator is supported.";
2     }
3     description
4         "Contains configuration nodes for the Authenticator PAE
5         associated with each port.";
6     leaf quiet-period {
7         type uint16;
8         units seconds;
9         default "60";
10        description
11            "Number of seconds that the authenticator remains in the quiet
12            state following a failed authentication exchange with the
13            supplicant.";
14        reference
15            "IEEE 802.1X-2020 Clause 8.6, Figure 12-3";
16    }
17    leaf reauth-period {
18        type uint32;
19        units seconds;
20        default "3600";
21        description
22            "This object indicates the time period of the
23            reauthentication to the supplicant.";
24        reference
25            "IEEE 802.1X-2020 Clause 8.6, Figure 12-3";
26    }
27    leaf reauth-enable {
28        type boolean;
29        default "false";
30        description
31            "Re-authentication is enabled or not.";
32        reference
33            "IEEE 802.1X-2020 Clause 5.8 and 8.9";
34    }
35    leaf retry-max {
36        type uint32;
37        default "2";
38        description
39            "Specifies the maximum number of re-authentication
40            attempts on an authenticator port before port is
41            unauthorized.";
42        reference
43            "IEEE 802.1X-2020 Clause 8.9";
44    }
45
46    leaf enabled {
47        type boolean;
48        config false;
49        description
50            "Set by PACP if the PAE can provide authentication. Will
51            be FALSE if the Port is not enabled, if the functionality
52            provided by the PAE is not available, or not implemented,
53            or the control variable enable has been cleared by
54            management, e.g. because the application scenario
55            authenticates a user and there is no user logged on.";
56        reference
57            "IEEE 802.1X-2020 Clause 8.4";
58    }
59    leaf authenticate {
60        type boolean;
61        config false;
62        description
63            "Set by the PAE client to request authentication, and
64            allows reauthentication while set. Cleared by the client
65            to revoke authentication. To enable authentication the
66            client also needs to clear failed (if set).";
67        reference
68            "IEEE 802.1X-2020 Clause 8.4";
69    }
70    leaf authenticated {
71        type boolean;
72        config false;
    
```



```

1      description
2          "Set by PACP if the PAE is currently authenticated, and
3          cleared if the authentication fails or is revoked.";
4      reference
5          "IEEE 802.1X-2020 Clause 8.4";
6  }
7  leaf failed {
8      type boolean;
9      config false;
10     description
11         "Set by PACP if the authentication has failed or has been
12         terminated. The cause could be a Fail returned by EAP,
13         either immediately or following a reauthentication, an
14         excessive number of attempts to authenticate (either
15         immediately or upon reauthentication), or the client
16         deasserting authenticate. The PACP will clear
17         authenticated as well as setting failed. Any ongoing
18         authentication exchange will be terminated (by the state
19         machines) if enable becomes FALSE and enabled will be
20         cleared, but failed will not be set.";
21     reference
22         "IEEE 802.1X-2020 Clause 8.4";
23 }
24 }
25
26 container kay {
27     when "../port-capabilities/mka = 'true'" {
28         description
29             "Applies when the MKA port capability is supported.";
30     }
31     description
32         "Contains configuration system level information for each
33         Interface supported by the KaY (Key Agreement Entity).";
34     leaf enable {
35         type boolean;
36         default "false";
37         description
38             "Set by management to enable (clear to disable) the use
39             of MKA.";
40         reference
41             "IEEE 802.1X-2020 Clause 9.16";
42     }
43     container actor {
44         description
45             "Contains configuration and operational nodes
46             associated with the actor";
47         leaf priority {
48             type uint8;
49             description
50                 "The Key Server Priority for all the ports actors.";
51             reference
52                 "IEEE 802.1X-2020 Clause 9.16";
53         }
54         leaf sci {
55             type dot1x-types:sci-list-entry;
56             config false;
57             description
58                 "The SCI assigned by the system to the port (applies
59                 to all the ports actors).";
60             reference
61                 "IEEE 802.1X-2020 Clause 9.16";
62         }
63     }
64     container key-server {
65         description
66             "Contains configuration and operational nodes
67             associated with the key
68             server.";
69         leaf priority {
70             type uint8;
71             description
72                 "The Key Server Priority for the Key Server for the

```

```

1         principal actor. Matches the actorPriority if the
2         actor is the Key Server";
3     reference
4         "IEEE 802.1X-2020 Clause 9.16";
5 }
6 leaf sci {
7     type dot1x-types:sci-list-entry;
8     config false;
9     description
10        "The SCI for Key Server for the principal actor. Null
11        if there is no principal actor, or that actor has no
12        live peers. Matches the actorSCI if the actor is the
13        Key Server.";
14    reference
15        "IEEE 802.1X-2020 Clause 9.16";
16 }
17 }
18 container group {
19     description
20        "Contains configuration nodes associated with the
21        group.";
22     leaf join {
23         type boolean;
24         default "true";
25         description
26            "Set if the KaY will accept Group CAKs distributed by
27            MKA.";
28         reference
29            "IEEE 802.1X-2020 Clause 9.16";
30     }
31     leaf form {
32         type boolean;
33         default "false";
34         description
35            "Set if the KaY will attempt to use point-to-point CAs
36            to distribute a Group CAK, if its principal actor is
37            the Key Server for all the point-to-point CAs.";
38         reference
39            "IEEE 802.1X-2020 Clause 9.16";
40     }
41     leaf new {
42         type boolean;
43         default "false";
44         description
45            "Set by management if a new Group CAK is to be
46            distributed, if the principal actor is the Key Server
47            for all point-to-point CAs. Cleared by the KaY when
48            distribution is complete.";
49         reference
50            "IEEE 802.1X-2020 Clause 9.16";
51     }
52 }
53
54 container macsec {
55     when "../port-capabilities/macsec = 'true'" {
56         description
57            "Applies when the MACsec port capability is
58            supported.";
59     }
60     description
61        "Contains configuration and operational nodes
62        associated with macsec.";
63     leaf capable {
64         type boolean;
65         description
66            "Set if MACsec is implemented.";
67         reference
68            "IEEE 802.1X-2020 Clause 9.16";
69     }
70     leaf desired {
71         type boolean;
72         default "true";

```

```

1      description
2      "Set if the participant desires MACsec frame protection.";
3      reference
4      "IEEE 802.1X-2020 Clause 9.16";
5  }
6
7  leaf protect {
8      type boolean;
9      config false;
10     description
11     "As used by the CP state machine, see 12.4.";
12     reference
13     "IEEE 802.1X-2020 Clause 9.16";
14 }
15 leaf validate {
16     type boolean;
17     config false;
18     description
19     "As used by the CP state machine, see 12.4.";
20     reference
21     "IEEE 802.1X-2020 Clause 9.16";
22 }
23 leaf replay-protect {
24     type boolean;
25     config false;
26     description
27     "As used by the CP state machine, see 12.4.";
28     reference
29     "IEEE 802.1X-2020 Clause 9.16";
30 }
31 }
32 leaf suspend-on-request {
33     type boolean;
34     default "true";
35     description
36     "Set by management to allow the KaYs principal actor to
37     initiate a suspension if it is the Key Server and another
38     participant has requested a suspension.";
39 }
40 leaf suspend-for {
41     type uint8;
42     default "0";
43     description
44     "Set by management to a non-zero number of seconds
45     between 1 and MKA Suspension Limit to initiate a
46     suspension (9.18) of that duration (if the KaYs principal
47     actor is the Key Server) or to request a suspension
48     (otherwise).";
49     reference
50     "IEEE 802.1X-2020 Clause 9.18";
51 }
52
53 leaf suspended-while {
54     type uint8;
55     config false;
56     description
57     "Read by management to determine if a suspension is in
58     progress and (when available) to discover the remaining
59     duration of that suspension";
60     reference
61     "IEEE 802.1X-2020 Clause 9.18";
62 }
63 leaf active {
64     type boolean;
65     config false;
66     description
67     "Set if there is at least one active actor, transmitting
68     MKPDUs.";
69     reference
70     "IEEE 802.1X-2020 Clause 9.16";
71 }
72 leaf authenticated {

```

```

1      type boolean;
2      config false;
3      description
4          "Set if the principal actor, i.e. the participant that
5          has the highest priority Key Server and one or more live
6          peers, has determined that Controlled Port communication
7          should proceed without MACsec.";
8      reference
9          "IEEE 802.1X-2020 Clause 9.16";
10     }
11     leaf secured {
12         type boolean;
13         config false;
14         description
15             "Set if the principal actor has determined that
16             communication should use MACsec.";
17         reference
18             "IEEE 802.1X-2020 Clause 9.16";
19     }
20     leaf failed {
21         type boolean;
22         config false;
23         description
24             "Cleared when authenticated or secured are set, set if
25             the latter are clear and MKA Life Time has elapsed since
26             an MKA participant was last created.";
27         reference
28             "IEEE 802.1X-2020 Clause 9.16";
29     }
30     container key-number {
31         config false;
32         description
33             "Contains operation state nodes for Key Numbers.";
34         leaf tx {
35             type dot1x-types:mka-kn;
36             description
37                 "The Key Number assigned by the Key Server to the SAK
38                 currently being used for transmission. Null if MACsec
39                 is not being used.";
40             reference
41                 "IEEE 802.1X-2020 Clause 9.16";
42         }
43         leaf rx {
44             type dot1x-types:mka-kn;
45             description
46                 "The Key Number assigned by the Key Server to the
47                 oldest SAK currently being used for reception. The same
48                 as txKN if a single SAK is currently in use (as will
49                 most often be the case). Null if MACsec is not being
50                 used.";
51             reference
52                 "IEEE 802.1X-2020 Clause 9.16";
53         }
54     }
55     container association-number {
56         config false;
57         description
58             "Contains operation state nodes for Association
59             Numbers.";
60         leaf tx {
61             type dot1x-types:mka-an;
62             description
63                 "The Association Number assigned by the Key Server for
64                 use with txKN. Zero if MACsec is not in use.";
65             reference
66                 "IEEE 802.1X-2020 Clause 9.16";
67         }
68         leaf rx {
69             type dot1x-types:mka-an;
70             description
71                 "The Association Number assigned by the Key Server for
72                 use with rxKN. The same as txAN if a single SAK is

```

```

1         currently in use. Zero if MACsec is not in use.";
2     reference
3         "IEEE 802.1X-2020 Clause 9.16";
4     }
5 }
6
7 list participants {
8     key "participant";
9     description
10        "Contains list of configuration and operational nodes
11        for each MKA participant supported by the KaY MKA
12        entity.";
13     leaf participant {
14         type uint32;
15         description
16             "Key into Participants list.";
17     }
18     leaf cached {
19         type boolean;
20         description
21             "Set by the KaY if the participants parameters are
22             cached. If set, cached can be cleared by management to
23             remove the participant from the cache.";
24     }
25     leaf active {
26         type boolean;
27         default "false";
28         description
29             "Set if the participant is active, i.e., is currently
30             transmitting periodic MKPDUs.";
31         reference
32             "IEEE 802.1X-2020 Clause 9.16";
33     }
34     leaf retain {
35         type boolean;
36         default "false";
37         description
38             "Set by management to retain the participant in the
39             cache, even if the KaY would normally remove it (due to
40             lack of use for example).";
41         reference
42             "IEEE 802.1X-2020 Clause 9.16";
43     }
44     leaf activate {
45         type enumeration {
46             enum default {
47                 description
48                     "The participant is from cached entries created by
49                     the KaY as part of normal operation, without
50                     explicit management, and is activated according to
51                     the implementation dependent policies of the KaY.";
52             }
53             enum disabled {
54                 description
55                     "The participant allows the cache information to be
56                     retained, but disabled for indefinite period.";
57             }
58             enum on-oper-up {
59                 description
60                     "Causing the participant to be activated when the
61                     PAEs part is activated, and therefore when the SecY
62                     or PACs Common Port becomes operational.";
63             }
64             enum always {
65                 description
66                     "Causing the participant to remain active all the
67                     time, even in the continued absence of partners.";
68             }
69         }
70         default "default";
71         description
72             "Controls when the participant is activated. Cached

```

```

1         entries created by the KaY as part of normal operation,
2         without explicit management, have the value Default,
3         and are activated according to the implementation
4         dependent policies of the KaY. This variable can be
5         set to any of its values by management. Disabled allows
6         the cache entry to be retained, but disabled for an
7         indefinite period. OnOperUp causes the participant to
8         be activated when the PAEs port (and therefore when the
9         SecY or PACs Common Port becomes MAC_Operational).
10        Always causes the participant to remain active all the
11        time, even in the continued absence of partners. If the
12        value is changed to Disabled or OnOperUp, the
13        participant ceases operation immediately and receipt of
14        MKPDUs with a matching CKN during a subsequent period
15        of twice MKA Life Time will not cause the participant
16        to become active once more.";
17        reference
18        "IEEE 802.1X-2020 Clause 9.16";
19    }
20
21    container peers {
22        config false;
23        description
24        "Contains operational state nodes associated with the
25        Peers.";
26        leaf-list live {
27            type dot1x-types:sci-list-entry;
28            description
29            "A list of the SCIs of the participants live
30            peers.";
31            reference
32            "IEEE 802.1X-2020 Clause 9.16";
33        }
34        leaf-list potential {
35            type dot1x-types:sci-list-entry;
36            description
37            "A list of the SCIs of the participants potential
38            peers.";
39            reference
40            "IEEE 802.1X-2020 Clause 9.16";
41        }
42    }
43    leaf ckn {
44        type dot1x-types:pae-ckn;
45        config false;
46        description
47        "The secure Connectivity Association Key Name for the
48        participant.";
49        reference
50        "IEEE 802.1X-2020 Clause 9.16";
51    }
52    leaf kmd {
53        type dot1x-types:pae-kmd;
54        config false;
55        description
56        "The Key Management Domain for the participant.";
57        reference
58        "IEEE 802.1X-2020 Clause 9.16";
59    }
60    leaf nid {
61        type dot1x-types:pae-nid;
62        config false;
63        description
64        "The NID for the participant.";
65        reference
66        "IEEE 802.1X-2020 Clause 9.16";
67    }
68    leaf auth-data {
69        type dot1x-types:pae-auth-data;
70        config false;
71        description
72        "Authorization data associated with the secure

```

```

1         Connectivity Association Key.";
2     reference
3         "IEEE 802.1X-2020 Clause 9.16";
4     }
5     leaf principal {
6         type boolean;
7         config false;
8         description
9             "Set if the participant is currently the principal
10            actor.";
11        reference
12            "IEEE 802.1X-2020 Clause 9.16";
13    }
14    leaf dist-ckn {
15        type dot1x-types:pae-ckn;
16        config false;
17        description
18            "The CKN for the last CAK distributed (either by the
19            actor or one of its partners). Null if this participant
20            has not been used to distribute a CAK.";
21        reference
22            "IEEE 802.1X-2020 Clause 9.16";
23    }
24    }
25    }
26
27    container logon-nid {
28        description
29            "Contains the configuration and operational related NID
30            information for the Logon Process. The Logon Process may
31            use Network Identifiers (NIDs) to manage its use of
32            authentication credentials, cached CAKs, and
33            announcements.";
34        leaf selected {
35            type dot1x-types:pae-nid;
36            description
37                "The NID currently configured for use by an access
38                controlled port when transmitting EAPOL-Start frames.
39                Defaults to the null NID.";
40            reference
41                "IEEE 802.1X-2020 Clause 12.5";
42        }
43        uses nid-group;
44
45        leaf connected {
46            type dot1x-types:pae-nid;
47            config false;
48            description
49                "The NID associated with the current connectivity
50                (possibly unauthenticated) provided by the operation of
51                the CP state machine.";
52            reference
53                "IEEE 802.1X-2020 Clause 12.5";
54        }
55        leaf requested {
56            type dot1x-types:pae-nid;
57            config false;
58            description
59                "The NID marked as Access requested in announcements, as
60                determined from EAPOL-Start frames. Defaults to the
61                selectedNID.";
62            reference
63                "IEEE 802.1X-2020 Clause 12.5";
64        }
65    }
66
67    container announcer {
68        when "../port-capabilities/announcements = 'true'" {
69            description
70                "Applies when the Announcements port capabilities are
71                supported.";
72        }

```

```

1      description
2          "Contains the configuration related Announcer
3          information.";
4      leaf enable {
5          type boolean;
6          default "false";
7          description
8              "A boolean indicating if the announcer is enabled or
9              not.";
10         reference
11             "IEEE 802.1X-2020 Clause 10.4";
12     }
13     list announce {
14         key "announces";
15         description
16             "Contains the configuration related status information
17             that the Announcers announce in the network announcement
18             of the PAE system.";
19         leaf announces {
20             type uint32;
21             description
22                 "Key into Announce list.";
23         }
24         uses nid-group;
25
26         leaf nid {
27             type dot1x-types:paе-nid;
28             config false;
29             description
30                 "The NID information to identify a received network
31                 announcement for the PAE.";
32             reference
33                 "IEEE 802.1X-2020 Clause 10.4";
34         }
35         leaf access-status {
36             type dot1x-types:paе-access-status;
37             config false;
38             description
39                 "Access Status reflects connectivity as a result of
40                 authentication attempts, and might be set directly by
41                 the system or configured by AAA protocols.";
42             reference
43                 "IEEE 802.1X-2020 Clause 10.4, Clause 12.5";
44         }
45     }
46 }
47
48 container listener {
49     when "../port-capabilities/listener = 'true'" {
50         description
51             "Applies when the Listener port capability is
52             supported.";
53     }
54     description
55         "Contains the configuration and operational Listener
56         node related information.";
57     leaf enable {
58         type boolean;
59         default "false";
60         description
61             "A boolean indicating if the listener is enabled or
62             not.";
63         reference
64             "IEEE 802.1X-2020 Clause 10.4";
65     }
66
67     list announcement {
68         key "announcements";
69         config false;
70         description
71             "A list containing the operational status information
72             that the Listeners receive in the network announcement of

```



```

1      the PAE system.";
2  leaf announcements {
3      type uint32;
4      description
5          "The key into the list of Announce nodes.";
6  }
7  leaf nid {
8      type dot1x-types:paе-nid;
9      description
10         "The NID information to identify a received network
11         announcement for the PAE.";
12     reference
13         "IEEE 802.1X-2020 Clause 10.4";
14 }
15 leaf kmd {
16     type dot1x-types:paе-kmd;
17     description
18         "The KMD information for this received network
19         announcement of the PAE.";
20     reference
21         "IEEE 802.1X-2020 Clause 10.4";
22 }
23 leaf specific {
24     type boolean;
25     description
26         "This object indicates the received announcement
27         information was specific to the receiving PAE, not
28         generic for all systems attached to the LAN.";
29     reference
30         "IEEE 802.1X-2020 Clause 10.4";
31 }
32 leaf access-status {
33     type dot1x-types:paе-access-status;
34     description
35         "The object information reflects connectivity as a
36         result of authentication attempts for this received
37         network announcement of the PAE.";
38     reference
39         "IEEE 802.1X-2020 Clause 10.4";
40 }
41 leaf requested-nid {
42     type boolean;
43     description
44         "The authenticated access has been requested for this
45         particular NID or not.";
46     reference
47         "IEEE 802.1X-2020 Clause 10.4";
48 }
49 leaf unauthenticated-access {
50     type dot1x-types:paе-access-status;
51     description
52         "The access capability of the ports clients without
53         authentication in this received network announcement of
54         the PAE";
55     reference
56         "IEEE 802.1X-2020 Clause 10.4";
57 }
58 leaf access-capabilities {
59     type dot1x-types:paе-nid-capabilities;
60     description
61         "The authentication and protection capabilities
62         supported for the NID.";
63     reference
64         "IEEE 802.1X-2020 Clause 10.4";
65 }
66 list cipher-suites {
67     key "index";
68     description
69         "A table contains the Cipher Suites information that
70         the Listeners receive in the network announcement of
71         the PAE system.";
72     reference

```

```

1         "IEEE 802.1X-2020 Clause 10.4";
2     leaf index {
3         type uint16;
4         description
5             "Key into cipher suite entry.";
6     }
7     leaf cipherSuite {
8         type string;
9         description
10            "cipher Suite identifier.";
11    }
12    leaf cipherSuiteCapability {
13        type uint32;
14        description
15            "Cipher Suite capability.";
16    }
17    }
18    }
19    }
20
21    container eapol-statistics {
22        config false;
23        description
24            "Contains operational EAPOL statistics.";
25        leaf invalid-eapol-frame-rx {
26            when "../port-type = 'real-port'" {
27                description
28                    "Applies when port is Real Port.";
29            }
30            type yang:counter32;
31            description
32                "The number of invalid EAPOL frames of any type that
33                have been received by this PAE.";
34            reference
35                "IEEE 802.1X-2020 Clause 12.8.1";
36        }
37        leaf eap-length-error-frames-rx {
38            when "../port-type = 'real-port'" {
39                description
40                    "Applies when port is Real Port.";
41            }
42            type yang:counter32;
43            description
44                "The number of EAPOL frames that the Packet Body Length
45                does not match a Packet Body that is contained within the
46                octets of the received EAPOL MPDU in this PAE.";
47            reference
48                "IEEE 802.1X-2020 Clause 12.8.1";
49        }
50        leaf eapol-announcements-rx {
51            when "../port-type = 'real-port'" {
52                description
53                    "Applies when port is Real Port.";
54            }
55            type yang:counter32;
56            description
57                "The number of EAPOL-Announcement frames that have been
58                received by this PAE";
59            reference
60                "IEEE 802.1X-2020 Clause 12.8.1";
61        }
62        leaf eapol-announce-reqs-rx {
63            when "../port-type = 'real-port'" {
64                description
65                    "Applies when port is Real Port.";
66            }
67            type yang:counter32;
68            description
69                "The number of EAPOL-Announcement-Req frames that have
70                been received by this PAE.";
71            reference
72                "IEEE 802.1X-2020 Clause 12.8.1";

```

```

1      }
2      leaf eapol-port-unavailable {
3          when "../../../port-type = 'real-port' and
4              ../../../port-capabilities/virtual-ports = 'true'" {
5              description
6                  "Applies when port is Real Port and when the virtual
7                  ports capability is supported.";
8          }
9          type yang:counter32;
10         description
11             "The number of EAPOL frames that are discarded because
12             their processing would require the creation of a virtual
13             port, for which there are inadequate or constrained
14             resources, or an existing virtual port and no such port
15             currently exists. If virtual port is not supported, this
16             object should be always 0.";
17         reference
18             "IEEE 802.1X-2020 Clause 12.8.1";
19     }
20     leaf eapol-start-frames-rx {
21         type yang:counter32;
22         description
23             "The number of EAPOL-Start frames that have been received
24             by this PAE.";
25         reference
26             "IEEE 802.1X-2020 Clause 12.8.1";
27     }
28     leaf eapol-eap-frames-rx {
29         type yang:counter32;
30         description
31             "The number of EAPOL-EAP frames that have been received
32             by this PAE.";
33         reference
34             "IEEE 802.1X-2020 Clause 12.8.1";
35     }
36     leaf eapol-logoff-frames-rx {
37         type yang:counter32;
38         description
39             "The number of EAPOL-Logoff frames that have been
40             received by this PAE.";
41         reference
42             "IEEE 802.1X-2020 Clause 12.8.1";
43     }
44     leaf eapol-mk-no-cfn {
45         type yang:counter32;
46         description
47             "The number of MKPDUs received with MKA not enabled or
48             CKN not recognized in this PAE.";
49         reference
50             "IEEE 802.1X-2020 Clause 12.8.1";
51     }
52     leaf eapol-mk-invalid-frames-rx {
53         type yang:counter32;
54         description
55             "The number of MKPDUs failing in message authentication
56             on receipt process in this PAE.";
57         reference
58             "IEEE 802.1X-2020 Clause 12.8.1";
59     }
60     leaf last-eapol-frame-source {
61         when "../../../port-type = 'real-port'" {
62             description
63                 "Applies when port is Real Port.";
64         }
65         type ieee:mac-address;
66         description
67             "The source MAC address of last received EAPOL frame by
68             this PAE.";
69         reference
70             "IEEE 802.1X-2020 Clause 12.8.2";
71     }
72     leaf last-eapol-frame-version {

```

```

1      type uint8;
2      description
3          "The version of last received EAPOL frame by this PAE.";
4      reference
5          "IEEE 802.1X-2020 Clause 12.8.2";
6  }
7  leaf eapol-supp-eap-frames-tx {
8      when "../port-type = 'real-port'" {
9          description
10             "Applies when port is Real Port.";
11     }
12     type yang:counter32;
13     description
14         "The number of EAPOL-EAP frames that have been
15         transmitted by the supplicant of this PAE.";
16     reference
17         "IEEE 802.1X-2020 Clause 12.8.3";
18 }
19 leaf eapol-logoff-frames-tx {
20     when "../port-type = 'real-port'" {
21         description
22             "Applies when port is Real Port.";
23     }
24     type yang:counter32;
25     description
26         "The number of EAPOL-Logoff frames that have been
27         transmitted by this PAE.";
28     reference
29         "IEEE 802.1X-2020 Clause 12.8.3";
30 }
31 leaf eapol-announcements-tx {
32     when "../port-type = 'real-port'" {
33         description
34             "Applies when port is Real Port.";
35     }
36     type yang:counter32;
37     description
38         "The number of EAPOL-Announcement frames that have been
39         transmitted by this PAE.";
40     reference
41         "IEEE 802.1X-2020 Clause 12.8.3";
42 }
43 leaf eapol-announce-reqs-tx {
44     when "../port-type = 'real-port'" {
45         description
46             "Applies when port is Real Port.";
47     }
48     type yang:counter32;
49     description
50         "The number of EAPOL-Announcement-Req frames that have
51         been transmitted by this PAE.";
52     reference
53         "IEEE 802.1X-2020 Clause 12.8.3";
54 }
55 leaf eapol-start-frames-tx {
56     type yang:counter32;
57     description
58         "The number of EAPOL-Start frames that have been
59         transmitted by this PAE.";
60     reference
61         "IEEE 802.1X-2020 Clause 12.8.3";
62 }
63 leaf eapol-auth-eap-frames-tx {
64     type yang:counter32;
65     description
66         "The number of EAPOL-EAP frames that have been
67         transmitted by the authenticator of this PAE.";
68     reference
69         "IEEE 802.1X-2020 Clause 12.8.3";
70 }
71 leaf eapol-mka-frames-tx {
72     type yang:counter32;

```

```

1      description
2      "The number of EAPOL-MKA frames with no CKN information
3      that have been transmitted by this PAE.";
4      reference
5      "IEEE 802.1X-2020 Clause 12.8.3";
6    }
7  }
8
9  container logon-process {
10   description
11   "Contains configuration and operational system level
12   information for each port to support the Logon Process(es)
13   status information.";
14   leaf logon {
15     type boolean;
16     default "false";
17     description
18     "A boolean indicating if the logon-process is enabled or
19     not.";
20     reference
21     "IEEE 802.1X-2020 Clause 12.5";
22   }
23
24   leaf connect {
25     type enumeration {
26       enum pending {
27         description
28         "Prevent connectivity by clearing the
29         controlledPortEnabled parameter.";
30       }
31       enum unauthenticated {
32         description
33         "Provide unsecured connectivity, setting
34         controlledPortEnabled.";
35       }
36       enum authenticated {
37         description
38         "Provide unsecured connectivity with authorization
39         data, setting controlledPortEnabled.";
40       }
41       enum secure {
42         description
43         "Provide secure connectivity, using SAKs provided by
44         the KaY (when available) and setting
45         controlledPortEnabled when those keys are installed
46         and in use, as specified in detail by the CP state
47         machine.";
48       }
49     }
50     config false;
51     description
52     "The Logon Process sets this variable to one of the
53     above values.";
54     reference
55     "IEEE 802.1X-2020 Clause 12.3";
56   }
57   leaf port-valid {
58     type boolean;
59     config false;
60     description
61     "Set if Controlled Port communication is secured as
62     specified by the MACsec control macsecProtect.";
63     reference
64     "IEEE 802.1X-2020 Clause 12.3";
65   }
66   list session-statistics {
67     key "session-id";
68     config false;
69     description
70     "Contains operational state nodes associated with the
71     session statistics.";
72     leaf session-id {

```

```

1      type dot1x-types:pae-session-id;
2      description
3          "Key into list of session statistics.";
4      reference
5          "IEEE 802.1X-2020 Clause 12.5.1";
6  }
7  leaf user-name {
8      type dot1x-types:pae-session-user-name;
9      description
10         "User name of the session.";
11     reference
12         "IEEE 802.1X-2020 Clause 12.5.1";
13 }
14 leaf octets-rx {
15     type yang:counter64;
16     description
17         "The number of octets received in this session of this
18         PAE.";
19     reference
20         "IEEE 802.1X-2020 Clause 12.5.1";
21 }
22 leaf octets-tx {
23     type yang:counter64;
24     description
25         "The number of octets transmitted in this session of
26         this PAE.";
27     reference
28         "IEEE 802.1X-2020 Clause 12.5.1";
29 }
30 leaf frames-rx {
31     type yang:counter64;
32     description
33         "The number of packets received in this session of
34         this PAE.";
35     reference
36         "IEEE 802.1X-2020 Clause 12.5.1";
37 }
38 leaf frames-tx {
39     type yang:counter64;
40     description
41         "The number of packets transmitted in this session of
42         this PAE.";
43     reference
44         "IEEE 802.1X-2020 Clause 12.5.1";
45 }
46 leaf time {
47     type uint32;
48     units "seconds";
49     description
50         "Session Time. The duration of the session in
51         seconds.";
52     reference
53         "IEEE 802.1X-2020 Clause 12.5.1";
54 }
55 leaf terminate-cause {
56     type enumeration {
57         enum common_port_MAC_operational_false {
58             description
59                 "Common Port for this PAE is not operational.";
60         }
61         enum system_access_control_disabled {
62             description
63                 "The system-access-control node of the pae-system
64                 is disabled or initialization process of this PAE
65                 is invoked.";
66         }
67         enum eapol_logoff_rx {
68             description
69                 "The PAE has received EAPOL-Logoff frame.";
70         }
71         enum eap_reauthentication_failure {
72             description

```

```
1         "EAP reauthentication has failed.";
2     }
3     enum mka-failure_termination {
4         description
5             "MKA failure or other MKA termination.";
6     }
7     enum new_session-beginning {
8         description
9             "New session beginning.";
10    }
11    enum not_terminated_yet {
12        description
13            "Not Terminated Yet.";
14    }
15    }
16    description
17        "The reason for the session termination.";
18    reference
19        "IEEE 802.1X-2020 Clause 12.5.1";
20 }
21 }
22 }
23 }
24 }
25
26 container nid-group {
27     description
28         "Contains both configuration and operational state nodes
29         associated with the PAE NID group.";
30     uses nid-group;
31 }
32
33 }
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